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Range of motion measurements: reference values and a database for comparison studies

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Summary. Many diseases and injuries can impair joint mobility. Normal reference values are needed to determine extent of impairment to assess and monitor joint motion. There is very little published data describing normal joint range of motion (ROM) for healthy men and women across a wide span of ages. We enrolled male and female subjects aged between 2 and 69 years who were free from conditions that could potentially limit joint mobility for the study. Nine licensed physical therapists used universal goniometers to determine passive joint motion bilaterally of elbow flexion, extension, supination and pronation, shoulder flexion, hip flexion and extension, knee flexion and extension, and ankle dorsiflexion and plantarflexion. Descriptive statistics were calculated for male and female subjects in four age groups: 2–8, 9–19, 20–44 and 45–69 years. Joint ROM measurements were

obtained on a total of 674 (53.6% female) healthy, normal subjects aged 2–69 years. Female subjects had greater joint mobility in all age groups in nearly all joints and the gender difference was most obvious in measures of ankle plantarflexion, elbow pronation and supination. Range of motion average values for all joints decreased with advancing age for both men and women and, in most cases, were significantly different than most commonly used normative values. Our study of ROM measurements taken by trained physical therapists on a large sample of healthy individuals revealed significant gender- and age-related variation that may be an important consideration in patient assessment.

Keywords: joint flexibility, joint range of motion, musculoskeletal system, reference values

Introduction

The International Classification of Functioning, Disability and Health (ICF) provides a framework describing relationships between body function and the ability to engage in life activities and recognizes joint mobility as an important component of health and a potential area of impairment [1]. Many diseases and injuries cause impaired joint mobility. Health care providers, including physicians and therapists, provide interventions

designed to restore joint mobility to reduce activity limitations. The severity of impaired joint mobility must be determined in comparison with normal reference values. Unfortunately, there are little published data describing normal joint mobility for healthy male and female subjects across a wide range of ages.

A common method of joint mobility assessment is the measurement of joint range of motion (ROM) using a universal goniometer. By aligning the stationary and movable arms of the device with specific bony landmarks on either side of the joint, the full extent of joint mobility can be measured in degrees. The most commonly used reference values for joint ROM are those published by the American Academy of Orthopaedic Surgeons (AAOS) [2]. The AAOS provides one or more reference values plus a range for each joint motion but the measures are not consistently stratified by age or gender.

Other studies on joint normal ROM have been reported but all have limitations in one or more of the

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following attributes: sample size, range of subject age, gender representation, the number of joints measured or the inconsistent methodology in measuring joint ROM [3–15].

The purpose of this study was to establish normative joint ROM values for upper and lower joints based upon healthy individuals of both genders across the lifespan using a standardized measurement approach. To accomplish this goal, we collected bilateral passive ROM measurements of the elbow, shoulder, hip, knee and ankle from a sample of individuals without known medical or physical conditions affecting the joint mobility. Our objective was to generate data that could be used to: (i) provide reference values for normal joint ROM for both genders and across the lifespan for comparison to people with haemophilia and other bleeding disorders, and (ii) provide a public database of joint ROM measures that could be used to assess impairment in joint mobility for other patient populations.

Materials and methods

Study design and subject enrollment

We used the published data on the statistical variability in ROM measurements taken on 109 normal men [9] to calculate the sample size needed for the study. As reported variability was different depending on the joint measured, the estimate was based on ankle plantarflexion as this measure showed intermediate variability. Based on these data, a sample size of 50 subjects would have 90% power to provide estimates of 'normal' ROM with a precision of $\pm 5\%$. To ensure adequate numbers of subjects representing both genders and a wide range of ages, enrollment was targeted to 50 male subjects and 50 female subjects in each of seven age groups (2–9, 10–19, ..., 60–69) for a total of 700 subjects.

This descriptive study determined measurements on a convenience sample of male and female volunteers aged 2–69 years recruited in a variety of settings such as community gatherings, schools, scientific meetings and workplaces. Study exclusions included conditions that could potentially limit joint mobility, as shown in Table 1. Weight and height were measured in light clothing. Body mass index (BMI) was calculated as weight in kilograms divided by the height in meters squared. Subjects with BMI >35 were excluded to minimize the possibility that joint ROM was limited because of excess adipose tissue. All subjects and parents of minor children gave informed consent to participate in the study, as appropriate.

Procedure

Before initiating the study, nine licensed physical therapists involved in data collection participated in a 1 day training to standardize procedures for measuring

Table 1. Subject Inclusion and Exclusion Criteria.

Inclusion criteria
• Male subjects and female subjects between 2 and 69 years of age
• Able to ambulate without assistance devices such as walkers, crutches or canes
Exclusion criteria
• Personal or family history of a bleeding disorder diagnosed by a physician
• Connective tissue disorder (e.g., rheumatoid arthritis, lupus, Ehler-Danlos or hypermobility syndromes, osteogenesis imperfecta) diagnosed by a physician or prescribed medications on a regular basis for joint pain
• History of any type of joint surgery including ligament repair or joint replacement
• Neurological diseases that may affect ROM (e.g., cerebral palsy, spina bifida, Parkinson's Disease, stroke, multiple sclerosis)
• Amputation of any part of any limb
• Acute joint injury including torn ligament, significant trauma, cast
• Recent injury including a bone fracture within the past 6 months or a joint sprain or dislocation within the past 3 months
• Pregnancy
• Diabetes mellitus diagnosed by a physician
• BMI >35
• Waist to Hip ratio >1

BMI, body mass index; ROM, range of motion.

joint ROM. After training, the participating therapists measured the joint ROM of a subset of 10 subjects. The differences in ROM measurements among the therapists were 5° or less.

Eleven joint motions in five joints were measured bilaterally producing a total of 22 ROM measurements for each subject. Subjects were positioned seated to measure elbow supination and pronation, as well as ankle dorsiflexion and plantar flexion. They were positioned supine to measure shoulder flexion, elbow flexion and extension, hip flexion, and knee flexion and extension. Subjects were positioned side lying to measure hip extension. The side-lying test position for hip extension was chosen to standardize procedures that could be applied to patients with hip flexion contractures or other joint mobility issues that make the standard prone test position difficult to assume. Each joint was moved passively to its full extent and end-point measurements were made to the nearest 1° using a universal goniometer.

Statistical methods

Age categories were grouped to encompass hormonal changes of prepuberty, reproductive and skeletal maturity and perimenopause that are known to affect joint laxity and body mass [16–20]. Based on these considerations and the results of exploratory analyses of the collected data, normative ROM values were calculated separately for male and female subjects in each of four age groups: (i) children (aged 2–8 years); (ii) adolescents (aged 9–19 years); (iii) young adults (aged 20–44 years); and (iv) older adults (aged 45–69 years).

Joint motions were expressed as means, standard deviations (SD), 95% confidence intervals (CI) and interquartile ranges (IQR). Paired *t*-tests were calculated

Table 2. Comparison of the demographic characteristics of male and female subjects.

	Female (361) N (%)	Male (313) N (%)	P-value*
Race			0.73
White	318 (88.1)	277 (88.5)	
African American	14 (3.9)	11 (3.5)	
Hispanic	16 (4.4)	10 (3.2)	
Other	13 (3.6)	15 (4.8)	
BMI (mean $\pm$ SD)	22.8 $\pm$ 4.5	23.8 $\pm$ 5.3	0.01
Age (years, mean $\pm$ SD)	34.7 $\pm$ 19.1	31.8 $\pm$ 19.7	0.06
Age categories (in years)			0.09
2–8	39 (10.8)	55 (17.6)	
9–19	56 (15.5)	48 (15.3)	
20–44	143 (39.6)	114 (36.4)	
45–69	123 (34.1)	96 (30.7)	

BMI, body mass index.

*The *P* values were calculated from chi-square test or Student *t*-test wherever appropriate.

to compare joint ROM values obtained from the right and left sides. Analysis of variance was used to assess differences in the mean joint ROM values by gender and across the four age groups. All analyses were carried out using the Statistical Analysis Software (SAS) version 9.1 (SAS Institute, Cary, NC, USA).

Results

From November 2003 to July 2006, joint ROM measurements were obtained on a total of 674 healthy and normal subjects. Subject demographics are shown in Table 2 and Fig. 1. Overall, subjects were predominately white (88.3%) and there were slightly more females

(54%) than males. Race and ethnicity was similar for both genders ($P = 0.73$). The average age of the study subjects was 33.3 years with female subjects marginally older than male subjects ($P = 0.06$). The average BMI of the population was 23.2, and was significantly higher in male subjects than in female subjects ($P = 0.009$). Lower BMI in female subjects was found primarily in subjects older than 19 years of age ($P = 0.0001$).

There were statistically significant differences between left and right sides in the measures of knee extension and flexion, shoulder flexion, and elbow extension and supination. However, the absolute differences in the means of left and right measures were very small ($<1^\circ$ in all cases). Therefore, for the remainder of the analyses, the averages of joint ROM measures from both sides were used as the joint ROM measurement for each subject.

For both male and female subjects, across all joints, mobility was greatest in the youngest age group and lowest in the oldest age group (Table 3). The greatest difference was seen in knee flexion with a 15° difference in mean ROM in the 2–8 years age group compared with 45–69 years age group. There was 10° or greater decrease from the youngest to the oldest age groups in mean ROM for ankle dorsiflexion and shoulder flexion in both genders and, in men, for hip flexion, ankle plantarflexion and hip extension. All of the other age-related joint ROM changes were $<10^\circ$ and there was very little difference among the age groups in measures of knee and elbow extension. More detailed univariate statistics are provided for all ROM measures in the Appendix.

Table 3. Normative values of joint range of motion in 674 normal subjects by gender and age.

Joint motion	Age 2–8*	Age 9–19*	Age 20–44*	Age 45–69*
Females (N)	39	56	143	123
Hip extension	26.2 (23.9–28.5)	20.5 (18.6–22.4)	18.1 (17.0–19.2)	16.7 (15.5–17.9)
Hip flexion	140.8 (139.2–142.4)	134.9 (133.0–136.8)	133.8 (132.5–135.1)	130.8 (129.2–132.4)
Knee flexion	152.6 (151.2–154.0)	142.3 (140.8–143.8)	141.9 (140.9–142.9)	137.8 (136.5–139.1)
Knee extension	5.4 (3.9–6.9)	2.4 (1.5–3.3)	1.6 (1.1–2.1)	1.2 (0.7–1.7)
Ankle dorsiflexion	24.8 (22.5–27.1)	17.3 (15.6–19.0)	13.8 (12.9–14.7)	11.6 (10.6–12.6)
Ankle plantar flexion	67.1 (64.8–69.4)	57.3 (54.8–59.8)	62.1 (60.6–63.6)	56.5 (55.0–58.0)
Shoulder flexion	178.6 (176.9–180.3)	171.8 (169.8–173.8)	172.0 (170.9–173.1)	168.1 (166.7–169.5)
Elbow flexion	152.9 (151.5–154.3)	149.7 (148.5–150.9)	150.0 (149.1–150.9)	148.3 (147.3–149.3)
Elbow extension	6.8 (5.2–8.4)	6.4 (4.7–8.1)	4.7 (3.9–5.5)	3.6 (2.6–4.6)
Elbow pronation	84.6 (82.8–86.4)	81.2 (79.6–82.8)	82.0 (81.0–83.0)	80.8 (79.7–81.9)
Elbow supination	93.7 (91.4–96.0)	90.0 (88.0–92.0)	90.6 (89.2–92.0)	87.2 (86.0–88.4)
Males (N)	55	48	114	96
Hip extension	28.3 (27.2–29.4)	18.2 (16.6–19.8)	17.4 (16.3–18.5)	13.5 (12.5–14.5)
Hip flexion	131.1 (129.4–132.8)	135.2 (133.0–137.4)	130.4 (129.0–131.8)	127.2 (125.7–128.7)
Knee flexion	147.8 (146.6–149.0)	142.2 (140.4–144.0)	137.7 (136.5–138.9)	132.9 (131.6–134.2)
Knee extension	1.6 (0.9–2.3)	1.8 (0.9–2.7)	1.0 (0.6–1.4)	0.5 (0.1–0.9)
Ankle dorsiflexion	22.8 (21.3–24.3)	16.3 (14.9–17.7)	12.7 (11.6–13.8)	11.9 (10.9–12.9)
Ankle plantar flexion	55.8 (54.4–57.2)	52.8 (50.8–54.8)	54.6 (53.2–56.0)	49.4 (47.7–51.1)
Shoulder flexion	177.8 (176.7–178.9)	170.9 (169.1–172.7)	168.8 (167.3–170.3)	164.0 (162.3–165.7)
Elbow flexion	151.4 (150.8–152.0)	148.3 (146.8–149.8)	144.6 (143.6–145.6)	143.5 (142.3–144.7)
Elbow extension	2.2 (0.9–3.5)	5.3 (3.6–7.0)	0.8 (0.1–1.5)	–0.7 (–1.5 to 0.1)
Elbow pronation	79.6 (78.8–80.4)	79.8 (77.8–81.8)	76.9 (75.6–78.2)	77.7 (76.5–78.9)
Elbow supination	86.4 (85.3–87.5)	87.8 (85.7–89.9)	85.0 (83.8–86.2)	82.4 (80.9–83.9)

*Data are presented as mean (95% confidence interval).

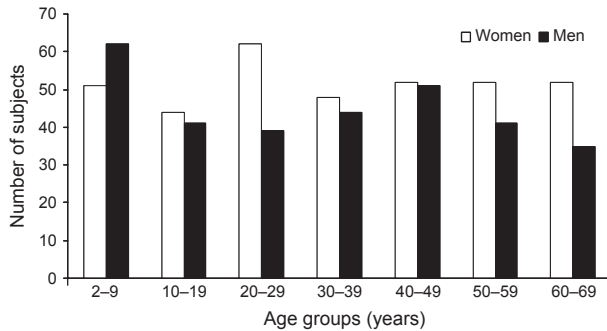


Fig. 1. Distribution of study subjects by age group and gender.

Discussion and conclusion

This article reports the largest study to date of joint ROM in healthy men and women across the lifespan. Although the NHANES I study incorporated a larger sample, that study as well as others was limited by the inclusion of subjects with self-reported joint pain and physician diagnosed musculoskeletal abnormalities [11]. In addition, comprehensive normative values were previously unavailable because reported studies included too few subjects, addressed joint ROM in a limited number of joints [4,6,7,11,13–15] or have included only one gender [3,6,9,13]. Active ROM may be decreased because of pain or weakness. Therefore, passive ROM better estimates actual joint motion. The current study, using passive ROM and excluding subjects with known joint pathology, found greater joint mobility in adults compared with earlier reports, and likely represents optimal joint mobility [11]. Previous reports have been inconsistent regarding the symmetry of joint ROM in young healthy individuals [4,6,9,13]. Although the differences in joint ROM measures that we obtained on the right and left sides of subjects were statistically significant, they are not clinically important, and support the use of a healthy limb for comparison with a limb being evaluated for potential alteration in joint ROM. However, our results provide evidence that one reference value for normal ROM is insufficient to characterize the gender differences and age-related changes in joint mobility in healthy people.

The general finding of this study that joint ROM tends to decline with advancing age is consistent with most other studies [7,9,11–13,21,22]. The greatest age-related difference was between children (aged 2–8 years) and all other groups. Where the 95% confidence intervals (95%CI) did not overlap, we can be relatively certain that there were real differences between the means of the various age groups. In most cases, the 95%CI for joint-specific ROM of 2–8 years age group did not overlap the confidence intervals for any of the other age groups. The 95%CI for the 9–18 years age group also generally differed from the

45–69 years age group but not from the 20–44 years age group. By contrast, although most mean ROM values of the 45–69 years age group were smaller than those of the 20–44 years age group, the 95%CI overlapped indicating that the differences were not statistically significant. Hip mobility in male subjects was an exception to our finding of relatively preserved joint mobility during adulthood. Mean hip extension ROM for male subjects in the 45–69 years age group was almost 4° less than that for male subjects in the 20–44 years age group with no overlap in the 95%CI.

Changes in joint ROM observed with puberty and ageing appear to be caused by changes in both joint laxity and body mass [16,17]. While increased BMI decreases joint mobility overall, in female subjects joint ROM increases with onset of puberty despite physiological increases in BMI [17]. The physiological increase in both BMI and joint ROM are influenced more by pubertal development than by age [23,24]. Indeed, joint laxity or hypermobility, defined as increased ROM relative to normals, and measured by the Beighton and Horan Joint Mobility Index, has been shown to increase in female subjects after entering puberty. However, this effect has not been shown in male subjects, who have joint laxity similar to that in female subjects prior to the onset of puberty [16]. Therefore, normal reference ranges in joint mobility must take into account both gender and pubertal status.

The age-related differences in joint mobility that we observed are also likely influenced by physiological changes in the musculoskeletal system that occur as part of normal ageing. Such changes include losses in the resilience of cartilage, decreased strength of skeletal muscle, reduced elasticity of ligaments and fat redistribution [25]. In addition, tissue changes associated with what ultimately result in osteoarthritis and osteoporosis may also have been present among our subjects, although we excluded individuals with a known diagnosis or significant symptoms of disease.

As the subjects in this study had no obvious joint pathology and the values represent passive rather than active ROM, the normal ROM values presented in this article represent optimal estimates of joint mobility for individuals in various age and gender groups. Such data should be useful to clinicians when assessing the impact of diseases such as haemophilia, rheumatoid arthritis or muscular dystrophy on joint mobility [26–30]. External normative values will be most useful for evaluation of disorders that have a symmetrical distribution as no healthy uninvolved limb is available for comparison. External ROM norms may also be useful for tracking the progression of the disease over time. A public use data set for use in comparative studies is available at the following web address: www.cdc.gov/normaljoints.

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Disclosures

The authors stated that they had no interests which might be perceived as posing a conflict or bias.

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Appendix

Table 1. Hip flexion range of motion in different gender and age groups.

Joint ROM	Age: 2–8 years		Age: 9–19 years		Age: 20–44 years		Age: 45–69 years	
	Female	Male	Female	Male	Female	Male	Female	Male
No. subjects	39	55	56	48	143	114	123	96
Mean	140.8	131.1	134.9	135.2	133.8	130.4	130.8	127.2
SD	5.1	6.3	7.4	7.6	7.9	7.5	8.8	7.4
Minimum	129	120	114.5	113	112	110	104	111
25th percentile	138.5	127.5	131	131	128.5	126.5	124.5	122.3
50th percentile	140.5	131	135.5	137.5	134.5	131	132	128
75th percentile	144.5	135	140.3	140.3	139.5	136	136	132.3
Maximum	154	144	150.5	149.5	151	144.5	154.5	143

ROM, range of motion.

Table 2. Hip extension range of motion in different gender and age groups.

Joint ROM	Age: 2–8 years		Age: 9–19 years		Age: 20–44 years		Age: 45–69 years	
	Female	Male	Female	Male	Female	Male	Female	Male
No. subjects	39	55	56	48	143	114	123	96
Mean	26.2	28.3	20.5	18.2	18.1	17.4	16.7	13.5
SD	7.4	4.0	7.1	5.6	6.6	5.9	7.0	5.2
Minimum	10.5	10.5	7.5	9	3	1	–10	3.5
25th percentile	22	28	13.8	14.3	13.5	13	11	10
50th percentile	25.5	30	20	17.3	17.5	17	16	12.5
75th percentile	31	30	26.8	21.5	22.5	21.5	21	17.5
Maximum	40	33.5	37	35.5	35	35.5	35	29

ROM, range of motion.

Table 3. Knee flexion range of motion in different gender and age groups.

Joint ROM	Age: 2–8 years		Age: 9–19 years		Age: 20–44 years		Age: 45–69 years	
	Female	Male	Female	Male	Female	Male	Female	Male
No. subjects	39	55	56	48	143	114	123	96
Mean	152.6	147.8	142.3	142.2	141.9	137.7	137.8	132.9
SD	4.5	4.7	5.7	6.3	6.3	6.5	7.4	6.3
Minimum	143	137.5	129	127.5	125	121.5	114.5	120
25th percentile	150	145	137.3	138	137.5	133	134	128.5
50th percentile	152	148.5	142	142	142	138.5	137.5	132.5
75th percentile	155.5	150	146.8	146.5	146.5	142	142.5	137.5
Maximum	162.5	162	156	156	159	152.5	160	150

ROM, range of motion.

Table 4. Knee extension range of motion in different gender and age groups.

Joint ROM	Age: 2–8 years		Age: 9–19 years		Age: 20–44 years		Age: 45–69 years	
	Female	Male	Female	Male	Female	Male	Female	Male
No. subjects	39	55	56	48	143	114	123	96
Mean	5.4	1.6	2.4	1.8	1.6	1.0	1.2	0.5
SD	4.9	2.5	3.5	3.2	2.8	2.3	2.9	1.9
Minimum	–1	–2.5	–3.5	–4.0	–6.5	–5.0	–7.0	–4.0

Table 4. Continued

Joint ROM	Age: 2–8 years		Age: 9–19 years		Age: 20–44 years		Age: 45–69 years	
	Female	Male	Female	Male	Female	Male	Female	Male
25th percentile	1.0	0	0	0	0	0	0	0
50th percentile	5.0	0	1.0	0	1.0	0	0	0
75th percentile	9.5	2.5	5.0	3.5	3.0	2.0	2.5	0.5
Maximum	18	10	13	12	10	11	10	8

ROM, range of motion.

Table 5. Ankle dorsiflexion range of motion in different gender and age groups.

Joint ROM	Age: 2–8 years		Age: 9–19 years		Age: 20–44 years		Age: 45–69 years	
	Female	Male	Female	Male	Female	Male	Female	Male
No. subjects	39	55	56	48	143	114	123	96
Mean	24.8	22.8	17.3	16.3	13.8	12.7	11.6	11.9
SD	7.2	5.6	6.4	5.1	5.5	5.9	5.5	5.0
Minimum	8	10	6	2.5	0.5	–0.5	–4	0
25th percentile	19	20	12.5	14	10.5	9	8	8.5
50th percentile	25.5	21.5	16	16	14	13	11.5	12
75th percentile	29	28.5	21.8	20	17.5	16	14.5	15
Maximum	41.5	32.5	33	30.5	36	27.5	25.5	23.5

ROM, range of motion.

Table 6. Ankle plantarflexion range of motion in different gender and age groups.

Joint ROM	Age: 2–8 years		Age: 9–19 years		Age: 20–44 years		Age: 45–69 years	
	Female	Male	Female	Male	Female	Male	Female	Male
No. subjects	39	55	56	48	143	114	123	96
Mean	67.1	55.8	57.3	52.8	62.1	54.6	56.5	49.4
SD	7.4	5.2	9.4	7.0	9.3	7.8	8.7	8.4
Minimum	52	49.5	42	34	44	37.5	37.5	30
25th percentile	61	52	50	48.8	54.5	50	50.5	43
50th percentile	67.5	55	54.5	52.8	62	55	55	49.5
75th percentile	72	58	63.3	58.5	69	59	62.5	54.3
Maximum	82	77.5	78	69	89.5	75	78	76

ROM, range of motion.

Table 7. Elbow flexion range of motion in different gender and age groups.

Joint ROM	Age: 2–8 years		Age: 9–19 years		Age: 20–44 years		Age: 45–69 years	
	Female	Male	Female	Male	Female	Male	Female	Male
No. subjects	39	55	56	48	143	114	123	96
Mean	152.9	151.4	149.7	148.3	150.0	144.6	148.3	143.5
SD	4.4	2.4	4.7	5.2	5.5	5.5	5.6	6.0
Minimum	145.0	146.5	140.5	137.5	135.0	129.5	135.0	122.5
25th percentile	150.0	150.0	146.5	145.3	146.0	141.5	145	139.8
50th percentile	151.5	150.5	150.0	147.5	150.0	144.5	149	143.5
75th percentile	156.0	152.0	152.5	151.8	153.0	147.5	151.5	147
Maximum	161.5	161.0	161.5	162.5	165.5	161.5	164	157.5

ROM, range of motion.

Table 8. Elbow extension range of motion in different gender and age groups.

Joint ROM	Age: 2–8 years		Age: 9–19 years		Age: 20–44 years		Age: 45–69 years	
	Female	Male	Female	Male	Female	Male	Female	Male
No. subjects	39	55	56	48	143	114	123	96
Mean	6.8	2.2	6.4	5.3	4.7	0.8	3.6	–0.7
SD	5.2	5.0	6.3	6.0	4.8	3.8	5.5	3.9
Minimum	–3.5	–10.5	–11.5	–12.5	–7.5	–8.5	–9.5	–12
25th percentile	3.0	0	1.8	1.5	0.5	–0.5	0	–3.0
50th percentile	7.0	0	6.8	3.5	4.0	0	3.0	0
75th percentile	9.0	3.5	10.3	10.0	8.5	3.0	7.0	0
Maximum	20.0	20.0	21.0	19.0	18.0	13	20	14.0

ROM, range of motion.

Table 9. Elbow pronation range of motion in different gender and age groups.

Joint ROM	Age: 2–8 years		Age: 9–19 years		Age: 20–44 years		Age: 45–69 years	
	Female	Male	Female	Male	Female	Male	Female	Male
No. subjects	39	55	56	48	143	114	123	96
Mean	84.6	79.6	81.2	79.8	82.0	76.9	80.8	77.7
SD	5.8	3.0	6.0	7.2	5.8	7.0	6.3	6.0
Minimum	71.0	70.0	59.5	45.5	67.0	60.0	56.5	45.0
25th percentile	80.0	79.0	78.3	77	78.0	73.5	78.0	74.5
50th percentile	84.0	80.0	81.8	81.0	82.0	77.5	81.0	79.0
75th percentile	89.5	81.0	85.0	83.5	86.0	82.0	85.0	81.5
Maximum	100.0	86.0	91.0	96.0	98.5	92.5	99.0	88.0

ROM, range of motion.

Table 10. Elbow supination range of motion in different gender and age groups.

Joint ROM	Age: 2–8 years		Age: 9–19 years		Age: 20–44 years		Age: 45–69 years	
	Female	Male	Female	Male	Female	Male	Female	Male
No. subjects	39	55	56	48	143	114	123	96
Mean	93.7	86.4	90.0	87.8	90.6	85.0	87.2	82.4
SD	7.3	4.2	7.5	7.3	8.5	6.6	6.9	7.5
Minimum	84.0	72.5	80.0	75.5	67.0	67.5	73.0	38.0
25th percentile	89.0	85.0	85.8	84.3	85.5	81.5	82.5	80.0
50th percentile	92.0	85.0	88.8	86.5	89.0	84.8	86.5	83.0
75th percentile	96.0	87.5	91.0	90.0	93.5	88.0	89.5	86.3
Maximum	115.0	105.5	122.5	111.5	124.5	117.0	109.0	101.5

ROM, range of motion.

Table 11. Shoulder flexion range of motion in different gender and age groups.

Joint ROM	Age: 2–8 years		Age: 9–19 years		Age: 20–44 years		Age: 45–69 years	
	Female	Male	Female	Male	Female	Male	Female	Male
No. subjects	39	55	56	48	143	114	123	96
Mean	178.6	177.8	171.8	170.9	172.0	168.8	168.1	164.0
SD	5.5	4.0	7.6	6.4	6.8	8.0	8.2	8.3
Minimum	162.5	163.5	154	157.5	147.5	151.5	147	130.5
25th percentile	175	175	166	166.3	167	162.5	163.5	159
50th percentile	180	180	170.5	170.8	172	167.8	168.5	164
75th percentile	181.5	180	180	176	178.5	176	173.5	169
Maximum	189	183	183	183	183	182.5	188	185.5

ROM, range of motion.